SAFETY DATA SHEET
(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 19-Jan-2024 **WAI2 - EGHS - EUROPEAN** **Revision Number** 7

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name Formic Acid (Acid Reagent)

Product No 182011
Unique Formula Identifier (UFI) Not applicable

REACH registration number Not applicable

Pure substance/mixture Mixture

Contains Formic Acid

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Fisher Scientific
Robert-Bosch-Str. 163505
Langenselbold, GERMANY
Tel.: +49 (6184) 90-6000

E-mail address wlp.techsupport@thermofisher.com

Made in USA

1.4. Emergency telephone number

24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION

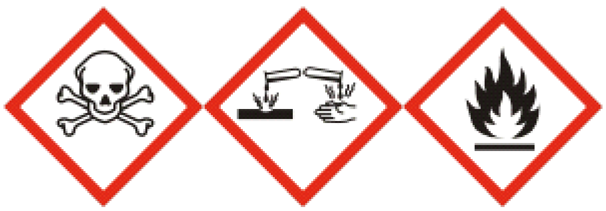
2.1. Classification of the substance or mixture
Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Acute Toxicity - Oral	Category 4 - (H302)
Acute toxicity - Inhalation (Vapors)	Category 3 - (H331)
Skin Corrosion/Irritation	Category 1 Sub-category B - (H314)
Serious eye damage/eye irritation	Category 1 - (H318)
Flammable liquids	Category 3 - (H226)

2.2. Label elements

Contains Formic Acid



Signal Word
Danger

Hazard Statements
H302 - Harmful if swallowed
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H226 - Flammable liquid and vapor

Precautionary Statements
P210 - Keep away from heat/sparks/open flames/hot surfaces. - No smoking
P264 - Wash face, hands and any exposed skin thoroughly after handling
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray
P270 - Do not eat, drink or smoke when using this product
P271 - Use only outdoors or in a well-ventilated area
P233 - Keep container tightly closed
P260 - Do not breathe dust/fume/gas/mist/vapors/spray
P243 - Take precautionary measures against static discharge
P240 - Ground/bond container and receiving equipment
P242 - Use non-sparking tools
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment
P303 + P361 + P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
P330 - Rinse mouth
P310 - Immediately call a POISON CENTER or doctor/physician
P311 - Call a POISON CENTER or doctor/physician
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P403 + P235 - Store in a well-ventilated place. Keep cool
P501 - Dispose of contents/ container to an approved waste disposal plant

2.3. Other hazards
No information available

General Hazards This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	EC No	CAS No	Weight %	CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567	REACH Reg. No
Formic Acid	EEC No. 200-579-1	64-18-6	80 - 90%	Flam. Liq. 3 (H226) Acute Tox. 4 (H302) Acute Tox. 3 (H331) Skin Corr. 1A (H314) Eye Dam. 1 (H318)	No information available
Water	EEC No. 231-791-2	7732-18-5	10 - 20%	Not classified	No information available

Component	CAS No	Specific concentration limits (SCL's)	M-Factor	Component notes
Formic Acid	64-18-6	Eye Irrit. 2 (H319) :: 2%<=C<10% Skin Corr. 1A (H314) :: C>=90% Skin Corr. 1B (H314) :: 10%<=C<90% Skin Irrit. 2 (H315) :: 2%<=C<10%	-	-
Water	7732-18-5	-	-	-

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Inhalation	If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Self-Protection of the First Aider	Use personal protective equipment as required. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	Causes burns by all exposure routes, Difficulty in breathing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
CO_2, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

Unsuitable Extinguishing Media
No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Environmental Precautions Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system. Vapors may accumulate to form explosive concentrations.

6.3. Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections
Refer to protective measures listed in Sections 7 and 8
See Section 8 for information on appropriate personal protective equipment
See Section 12 for additional Ecological Information
See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling
Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

General hygiene considerations
Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions
Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

7.3. Specific end use(s)

Specific Use(s)
Use as laboratory reagent

Risk Management Methods (RMM)
The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE -** 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

Component	European Union	The United Kingdom	France	Belgium	Spain
Formic Acid	TWA: 5 ppm (8hr) TWA: 9 mg/m³ (8hr)	STEL: 15 ppm 15 min STEL: 28.8 mg/m³ 15 min	TWA / VME: 5 ppm (8 heures). indicative limit TWA / VME: 9 mg/m³ (8	TWA: 5 ppm 8 uren TWA: 9.5 mg/m³ 8 uren STEL: 10 ppm 15	TWA / VLA-ED: 5 ppm (8 horas) TWA / VLA-ED: 9 mg/m³

		TWA: 5 ppm 8 hr TWA: 9.6 mg/m³ 8 hr	heures). indicative limit	minuten STEL: 19 mg/m³ 15 minuten	(8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Formic Acid	TWA: 5 ppm 8 ore. Time Weighted Average TWA: 9 mg/m³ 8 ore. Time Weighted Average	TWA: 5 ppm (8 Stunden). AGW - exposure factor 2 TWA: 9.5 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 5 ppm (8 Stunden). MAK TWA: 9.5 mg/m³ (8 Stunden). MAK Höhepunkt: 10 ppm Höhepunkt: 19 mg/m³	STEL: 10 ppm 15 minutos TWA: 5 ppm 8 horas TWA: 9 mg/m³ 8 horas	STEL: 5 mg/m³ 15 minuten	TWA: 3 ppm 8 tunteina TWA: 5 mg/m³ 8 tunteina STEL: 10 ppm 15 minuutteina STEL: 19 mg/m³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Formic Acid	MAK-KZGW: 5 ppm 15 Minuten MAK-KZGW: 9 mg/m³ 15 Minuten MAK-TMW: 5 ppm 8 Stunden MAK-TMW: 9 mg/m³ 8 Stunden Ceiling: 5 ppm Ceiling: 9 mg/m³	TWA: 5 ppm 8 timer TWA: 9 mg/m³ 8 timer STEL: 10 ppm 15 minutter STEL: 18 mg/m³ 15 minutter	STEL: 10 ppm 15 Minuten STEL: 19 mg/m³ 15 Minuten TWA: 5 ppm 8 Stunden TWA: 9.5 mg/m³ 8 Stunden	STEL: 15 mg/m³ 15 minutach TWA: 5 mg/m³ 8 godzinach	TWA: 5 ppm 8 timer TWA: 9 mg/m³ 8 timer STEL: 10 ppm 15 minutter. value calculated STEL: 18 mg/m³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Formic Acid	TWA: 5 ppm TWA: 9.0 mg/m³	TWA-GVI: 5 ppm 8 satima. TWA-GVI: 9 mg/m³ 8 satima.	TWA: 5 ppm 8 hr. TWA: 9 mg/m³ 8 hr. STEL: 15 ppm 15 min STEL: 27 mg/m³ 15 min	TWA: 5 ppm TWA: 9 mg/m³	TWA: 9 mg/m³ 8 hodinách. Ceiling: 18 mg/m³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Formic Acid	TWA: 5 ppm 8 tundides. TWA: 9 mg/m³ 8 tundides.	TWA: 5 ppm 8 hr TWA: 9 mg/m³ 8 hr	TWA: 5 ppm TWA: 9 mg/m³	TWA: 9 mg/m³ 8 órában. AK	TWA: 5 ppm 8 klukkustundum. TWA: 9 mg/m³ 8 klukkustundum. Skin notation Ceiling: 10 ppm Ceiling: 18 mg/m³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Formic Acid	TWA: 5 ppm TWA: 9 mg/m³	TWA: 5 ppm IPRD TWA: 9 mg/m³ IPRD	TWA: 5 ppm 8 Stunden TWA: 9 mg/m³ 8 Stunden	TWA: 5 ppm TWA: 9 mg/m³	TWA: 5 ppm 8 ore TWA: 9 mg/m³ 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Formic Acid	Skin notation MAC: 1 mg/m³	TWA: 5 ppm TWA: 9.0 mg/m³	TWA: 5 ppm 8 urah TWA: 9 mg/m³ 8 urah STEL: 10 ppm 15 minutah STEL: 18 mg/m³ 15 minutah	Indicative STEL: 5 ppm 15 minuter Indicative STEL: 9 mg/m³ 15 minuter TLV: 3 ppm 8 timmar. NGV TLV: 5 mg/m³ 8 timmar. NGV	TWA: 5 ppm 8 saat TWA: 9 mg/m³ 8 saat

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS70 General methods for sampling airborne gases and vapours
MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography
MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL)
See table for values

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Formic Acid 64-18-6 (80 - 90%)			DNEL = 9.5mg/m^3	

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Formic Acid 64-18-6 (80 - 90%)	PNEC = 2mg/L	PNEC = 13.4mg/kg sediment dw	PNEC = 1mg/L	PNEC = 7.2mg/L	PNEC = 1.5mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water intermittent	Food chain	Air
Formic Acid 64-18-6 (80 - 90%)	PNEC = 0.2mg/L	PNEC = 1.34mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location Ensure adequate ventilation, especially in confined areas Use explosion-proof electrical/ventilating/lighting equipment

Personal protective equipment

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur:. Goggles.

Skin and body protection Wear protective gloves/protective clothing.

Respiratory Protection No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Environmental exposure controls Prevent product from entering drains

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State Liquid
Appearance Clear
Odor pungent
Odor Threshold No information available
pH 2.3

PH Range 1.3 - 3.3

Property	Values	Remarks • Method
Melting point/freezing point	No information available	
Boiling Point/Range	> 100 °C / 212 °F	
Flash Point (High in °C)	60 °C / 140 °F	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties		
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity
No information available

10.2. Chemical stability
Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions
None under normal processing

10.4. Conditions to avoid
Keep away from open flames, hot surfaces and sources of ignition

10.5. Incompatible materials
No information available

10.6. Hazardous decomposition products
Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Acute Toxicity
Unknown Acute Toxicity 0 % of the mixture consists of ingredient(s) of unknown toxicity.
The following values are calculated based on chapter 3.1 of the GHS document
- **ATEmix (oral)** 1,294.00 mg/kg
- **ATEmix (inhalation-gas)** 824.00 ppm
- **ATEmix (inhalation-dust/mist)** 1.10 mg/L

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Formic Acid	LD50 = 1100 mg/kg (Rat)		LC50 = 7.85 mg/L (Rat) 4 h
Water	LD50 > 90 mL/kg (Rat)		

Skin Corrosion/Irritation Causes severe burns

(b) skin corrosion/irritation; Category 1. B.

Serious eye damage/eye irritation Risk of serious damage to eyes

Sensitization No information available

Mutagenic Effects No information available

Carcinogenic effects No information available

Reproductive Effects No information available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

Symptoms Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Aspiration hazard No information available

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity
Ecotoxicity effects Contains a substance which is:. Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

0% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Formic Acid	EC50: = 26.9 mg/L, 72h (Desmodesmus subspicatus) EC50: = 25 mg/L, 96h (Desmodesmus subspicatus)	-	EC50: 138 - 165.6 mg/L, 48h Static (Daphnia magna) EC50: = 120 mg/L, 48h (Daphnia magna)

12.2. Persistence and degradability

Persistence Persistence is unlikely.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Formic Acid	-1.9	0.22 dimensionless

12.4. Mobility in soil

Highly mobile in soils

Mobility

Will likely be mobile in the environment due to its water solubility

Component	log Pow
Formic Acid 64-18-6 (80 - 90%)	-1.9

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

Other Information Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1 UN-No UN1779
14.2 Proper Shipping Name FORMIC ACID SOLUTION
14.3 Hazard Class 8
14.4 Packing Group II

Description	UN1779, FORMIC ACID SOLUTION, 8 (3), II, (50°C C.C.)
14.5 Marine Pollutant	Not Applicable
14.6 Special Provisions	None
EmS No.	F-E, S-C
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	No information available

ADR

14.1. UN number	UN1779
14.2. UN proper shipping name	FORMIC ACID SOLUTION
14.3. Transport hazard class(es)	8
Subsidiary Hazard Class	3
14.4. Packing group	II

ICAO

14.1 UN-No	UN1779
14.2 Proper Shipping Name	FORMIC ACID SOLUTION
14.3 Hazard Class	8
14.4 Packing Group	II
Description	UN1779, FORMIC ACID SOLUTION, 8 (3), II
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

IATA

14.1 UN-No	UN1779
14.2 Proper Shipping Name	FORMIC ACID SOLUTION
14.3 Hazard Class	8
Subsidiary Hazard Class	3
14.4 Packing Group	II
Description	UN1779, FORMIC ACID SOLUTION, 8 (3), II
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None
ERG Code	8L

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS), U.S.A. (TSCA).

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Formic Acid	64-18-6	200-579-1	-	-	X	X	KE-17233	X	X
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Formic Acid	64-18-6	X	ACTIVE	X	-	X	X	X
Water	7732-18-5	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

European Union

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Formic Acid	64-18-6	-	Use restricted. See item 75. (see link for restriction details)	-
Water	7732-18-5	-	-	-

https://echa.europa.eu/substances-restricted-under-reach

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)
Formic Acid 64-18-6 (80 - 90%)	WGK1

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Formic Acid 64-18-6 (80 - 90%)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H302 - Harmful if swallowed
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage

H226 - Flammable liquid and vapor

Key or legend to abbreviations and acronyms used in the safety data sheet

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
TWA TWA (time-weighted average)
Ceiling Maximum limit value
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (Volatile Organic Compound)
STEL STEL (Short Term Exposure Limit)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Full text of H-Statements referred to under section 3
H226 - Flammable liquid and vapor
H302 - Harmful if swallowed
H331 - Toxic if inhaled
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.
Issue Date	No information available
Revision Date	19-Jan-2024
Reason for revision	SDS sections updated.

This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended. .

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End of Safety Data Sheet